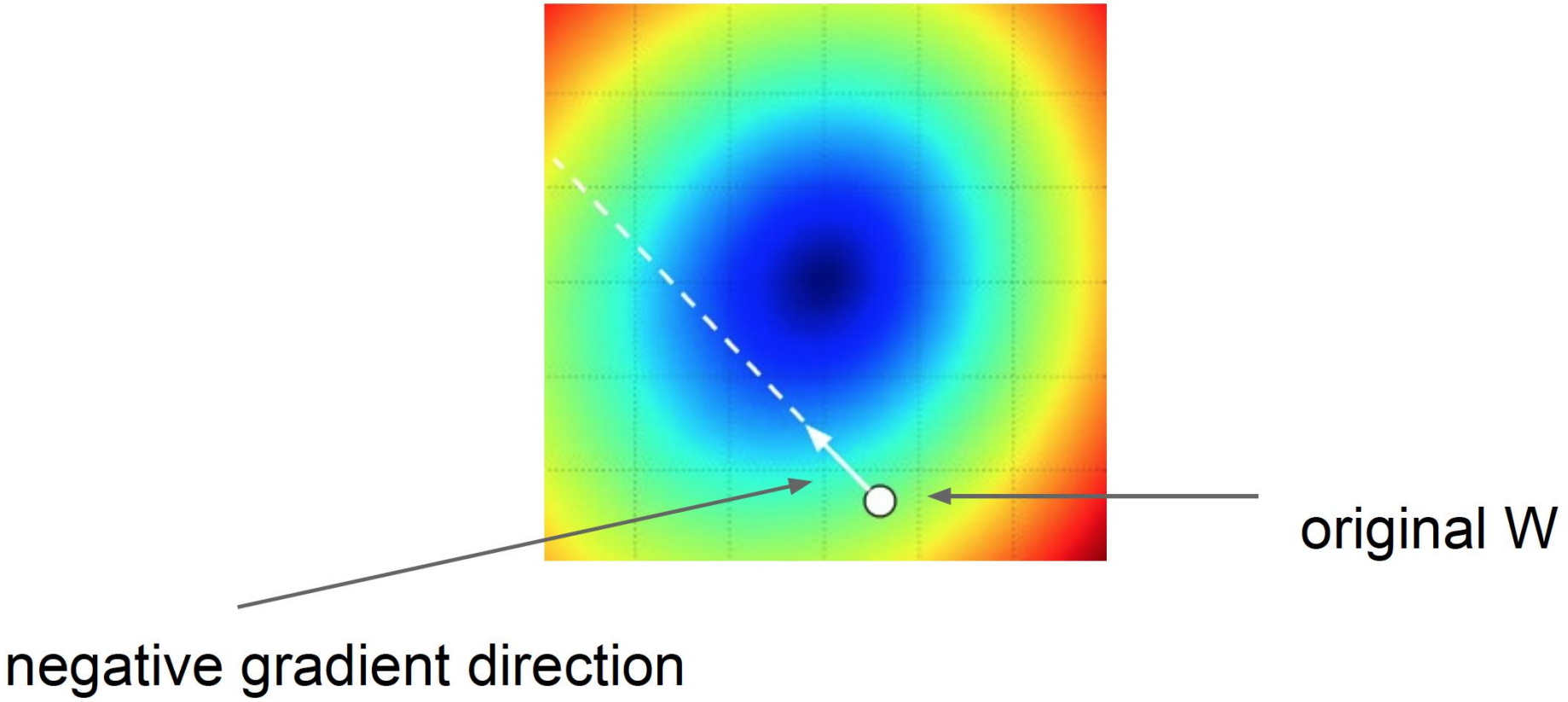
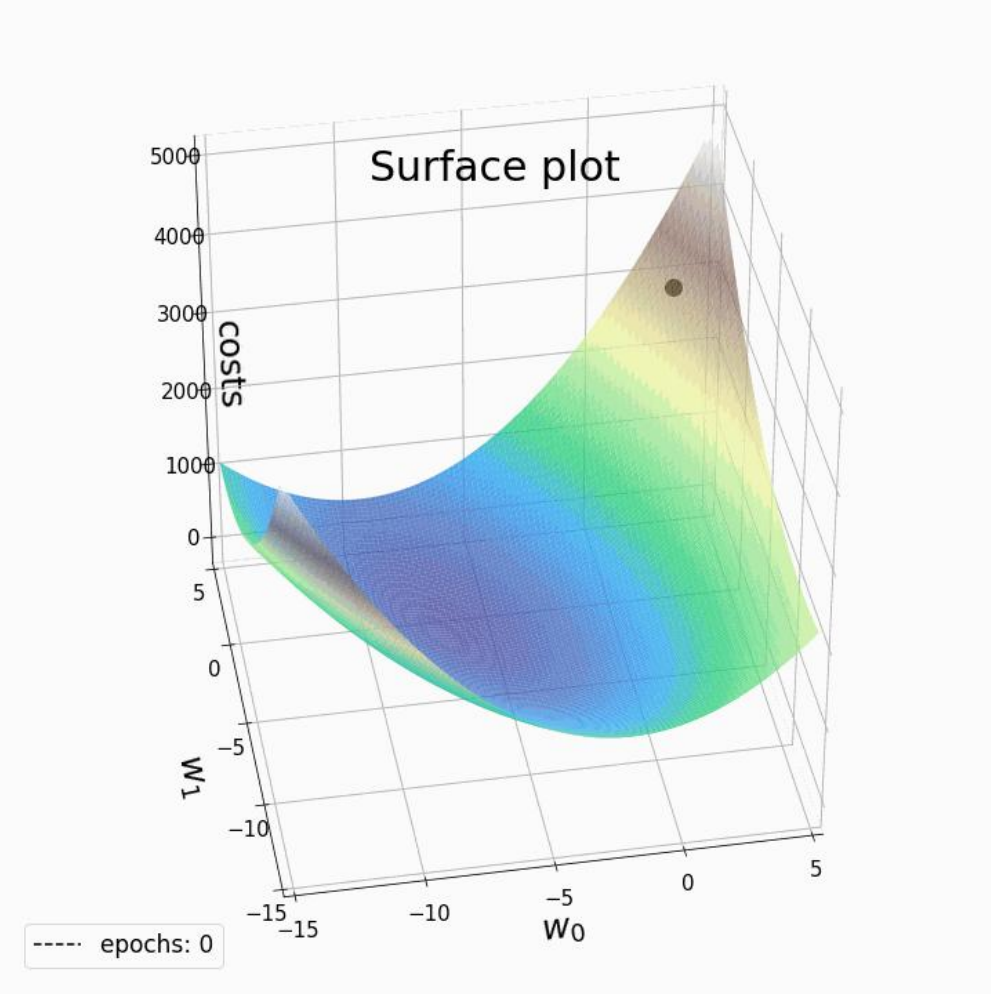


Gradient descent: walk in the direction opposite gradient

- **Q**: How far?
- **A**: Step size: *learning rate*
- Too big: will miss the minimum
- Too small: slow convergence

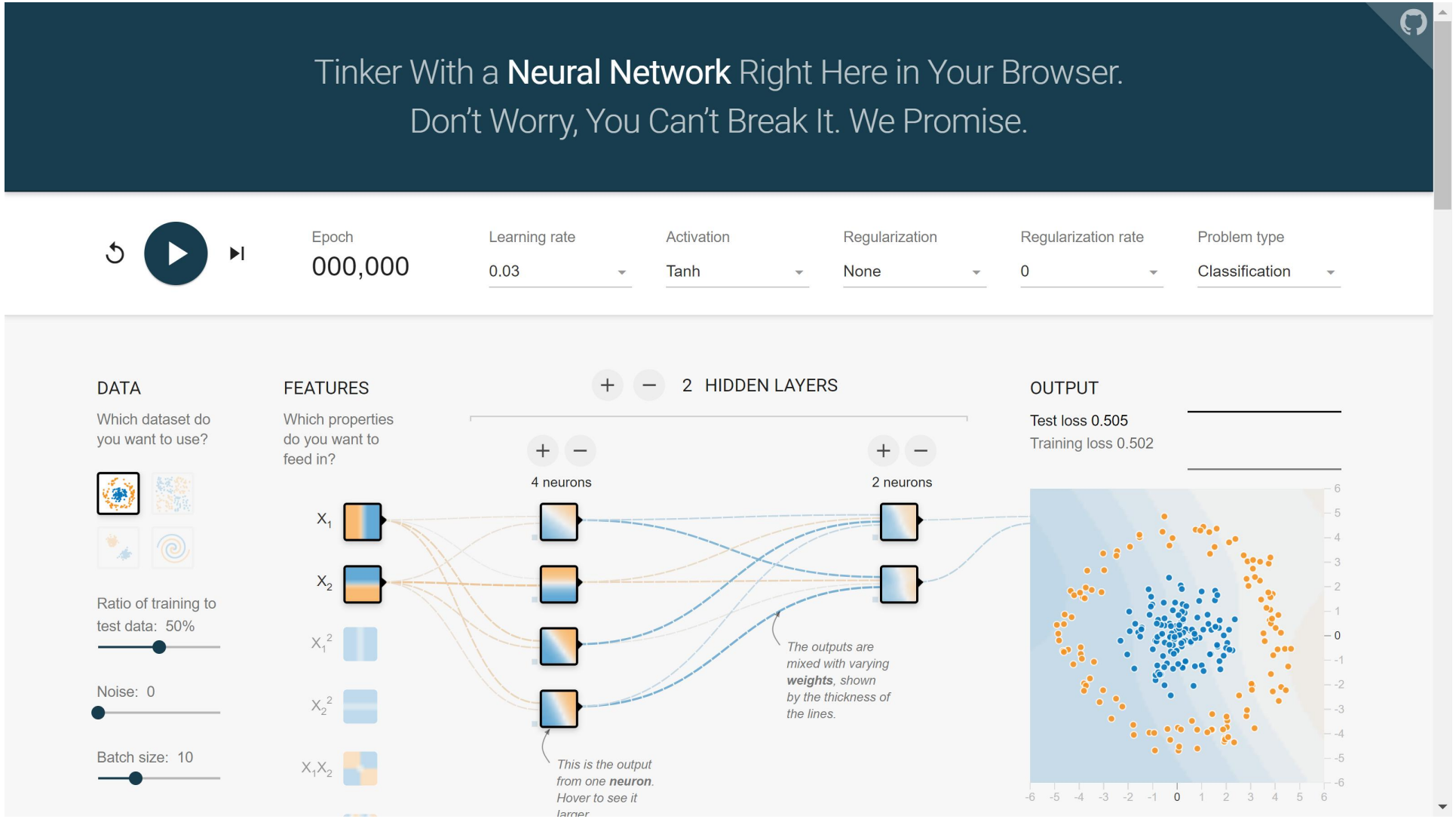
2D example of gradient descent



- In reality, in deep learning we are optimizing a highly complex loss function with millions of variables (or more)
- More on this later…

https://laptrinhx.com/gradient-descent-animation-2-multiple-linear-regression-3070246823/

2D example: TensorFlow Playground



https://playground.tensorflow.org

Questions?

Multi-layer perceptrons: just the beginning

- MLPs can work well with simple image classification problems (e.g., digit recognition)
- For more complex recognition problems (e.g., recognizing 1,000s of categories), we need more technology (convolutional neural networks and transformers)
- MLPs have a surprising application: 3D reconstruction!
  - See NeRFs (neural radiance fields) next time